Richard Morris, PhD

DATA SCIENTIST | CAUSAL INFERENCE | BEHAVIORAL & HEALTH

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PROFESSIONAL SUMMARY

Personable and pragmatic data scientist with 10+ years turning unstructured, real-world people data into causal answers for government, healthcare, and research organizations. Expert in observational and quasi-experimental causal inference, experimental design, longitudinal modeling, and large-scale survey science, with deep command of difference-in-differences, Bayesian estimation, and multilevel regression in R. Track record of driving analyses to secure multi-million dollar funding and health system decisions, translating research evidence into insights executives can act on. Ready to help people, product and health analytics teams turn quantitative and behavioral data into decisions grounded in sound methodology and research.

SKILLS & EXPERTISE

- **Causal Inference & Experimental Design:** Difference-in-Differences (DiD) | Instrumental Variables | Propensity Matching | Randomized & Quasi-Experimental Design | Bayesian Models
- **Programming & Tools:** R (tidyverse, brms, lme4) | Python (Pandas) | SQL
- **Survey & Measurement Science:** Survey Design & Programming (Qualtrics, REDCap) | Sampling-Bias Correction (Poststratification) | Psychometrics & Scale Validation
- **Communication & Delivery:** Stakeholder & Executive Reporting | Data Visualization & Dashboards (R-Shiny) | Client-Facing Consulting
- **Statistical & Longitudinal Modeling:** Longitudinal & Mixed-Effects Modeling | Regression & Statistical Inference | Predictive Modeling (LASSO)
- **Behavioral & Health Data:** Behavioral & Human-Data Analysis | Electronic Health Records (EHR) & Real-World Evidence (RWE) | Meta-Analysis & Evidence Synthesis

PROFESSIONAL EXPERIENCE

INDEPENDENT CONSULTANT, REMOTE

Senior Data Scientist & Research Consultant **Oct 2017 – Present**

Independent quantitative consultant for government, healthcare, and research clients, including an 8-year role with the University of Sydney, and a recent AI-model evaluation for Qeios.

- Designed a centered-difference generalized additive mixed model in R to separate genuine cohort effects from age and period trends across 20 years of national panel data (N=27,572), published in PNAS (2023).
- Built a cross-validated LASSO burnout-risk screening model in R (AUC 0.71) deployed to 129,000+ health-system workers, with 94% agreeing with the assigned risk rating, to target early intervention before crisis.
- Synthesized 81 digital health programs (98 apps, 25,500+ users) with network meta-analysis and Bayesian meta-regression in R, ranking which features drove engagement to prioritize an evidence-based feature roadmap.
- Produced population-representative workforce estimates from non-probability survey data using multilevel regression with poststratification in R, correcting sampling bias for national workplace-health policy analysis.

SPEED-EXTRACT, NSW MINISTRY OF HEALTH, SYDNEY, AUSTRALIA

Lead Data Analyst (Contract) **Dec 2018 – Mar 2020**

Led analysis for a feasibility test studying whether routine electronic health record (EHR) data could support clinical quality and safety monitoring at scale across 6 hospital sites.

- Designed a Bayesian validation study in R to determine how accurately hospital records identified cardiac events, showing routine coding missed 31% of one heart-attack subtype and supporting a $700M monitoring initiative.

- Validated and linked 5M+ electronic health records across 6 facilities against gold-standard clinical review, establishing data provenance with senior clinicians to certify the records for automated quality monitoring.
- Delivered 8 near-real-time clinical quality and safety indicators through automated R-Shiny dashboards, replacing manual reporting and presenting results to health-department leadership and the Minister of Health.

Ernst & Young (EY), Sydney, Australia

Data Scientist (Contract) **Oct 2017 – Dec 2019**

Owned the quantitative analysis for an independent evaluation of a federal early-psychosis program to deliver causal evidence to senior government stakeholders.

- Estimated an early-psychosis program's causal impact via Bayesian difference-in-differences (DiD), comparing treated participants against a matched comparison group, one of eight analyses informing a $200M funding decision.
- Translated the treatment-effect estimate into cost-effectiveness terms by integrating an economic loss function over the Bayesian posterior, giving government stakeholders a budget-ready figure rather than a p-value.
- Designed and deployed a national engagement survey to 3,000+ participants and mapped the participant journey with funnel and drop-off analysis, pinpointing where and why young people disengaged from the program.

Sydney Informatics Hub, University of Sydney, Sydney, Australia

Statistics Consultant **Jan 2017 – Aug 2017**

- Advised 20+ concurrent research projects on study design and causal-inference methods as part of a six-person consulting team, and introduced a JIRA project-tracking workflow that raised team throughput roughly 25%.

Brain & Mind Centre, University of Sydney, Sydney, Australia

Senior Research Scientist **Aug 2011 – Jan 2017**

Established and led a human decision research program across healthy and clinical populations, running controlled behavioral and brain-imaging experiments that adapted animal models of learning for humans.

- Built computational models of human causal learning from behavioral and brain-imaging data, identifying neural signals for goal-directed choice and publishing in Nature Communications, since adopted by other labs.

Education

PhD, Psychology
University of New South Wales (Sydney, Australia)

Postdoctoral Fellowship
University of Vermont (Burlington, VT)

MSc (Research), Quantitative Methods
University of Sydney (Sydney, Australia)

BA (Philosophy) / BSc Honours (Psychology)
University of New South Wales (Sydney, Australia)